

Chapter 1: Numerical Methods

We introduce a locally mass conserving numerical framework. We solve the static mechanics system with conforming mixed finite element method (MFEM), where the normal flux are captured at the edges of element by the MFEM scheme. The weak form is adjusted to be locally mass conservative. We solve the transport system with finite volume method (FVM), which is naturally a locally mass conservative scheme. We discuss MFEM and FVM separately and give error estimation. We provide a detailed coupling strategy combining MFEM, FVM and eutectic phase package in the end of this chapter.

1.1 Quadrilateral Mesh

We consider a bounded computational domain $\Omega \subset \mathbb{R}^2$. Here, $\partial\Omega$ denotes the Lipschitz-continuous boundary of the domain. Let $\mathcal{T}_h$ be a conforming mesh of quadrilaterals covering Ω . The quadrilateral $E \in \mathcal{T}_h$ is referred as element in finite element context, while referred as cell in finite volume context. Each quadrilateral E is assumed to be strictly convex, not degenerate into a triangle, line or point. For ease of implementation, we use a logically rectangular mesh, where each element(cell) is indexed using Cartesian coordinates $\{i, j\}$. We show a reference element(cell) $\hat{E} = [-1, 1]^2$, an unscaled element $\tilde{E}$ and a scaled general quadrilateral element(cell) $E/H = \tilde{E}$ in Figure 1.1. There exists a non-affine bijective and smooth map $\mathbf{F}_{\hat{E}} : \hat{E} \rightarrow \tilde{E}$ mapping the vertices of $\hat{E}$ to the vertices of $\tilde{E}$.

The edges of element E are denoted by e_i , $i \in \{0, 1, 2, 3\}$ in the counter-clockwise direction. We begin counting from zero, first to align with the indexing convention of C++, and second to facilitate representation in terms of modulo arithmetic, which also start from zero. The vertices are represented as $v_i = e_i \cap e_{i+1}$, and $i \equiv i \pmod{4}$. Here, $\boldsymbol{\nu}_i$ and $\boldsymbol{\tau}_i$ are used to represent the unit normal and the unit

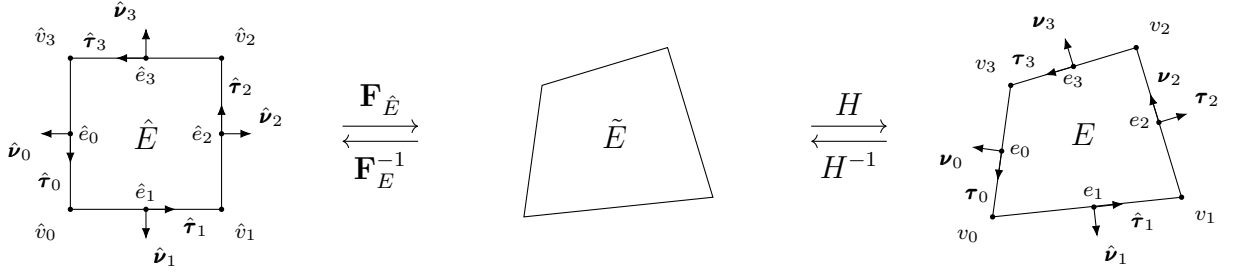


Figure 1.1: An exhibition of a reference square and general non-degenerate quadrilateral element(cell) before and after scaling.

tangent vector respectively. The unit normal vector $\boldsymbol{\nu}_i$ points outerwards the quadrilateral. The tangent $\boldsymbol{\tau}_i$ is generated by rotating $\boldsymbol{\nu}_i$ counterclockwise by right-hand rule.

In Figure 1.2, we illustrate a full logically rectangular quasi-uniform quadrilateral mesh $\mathcal{T}_h$, which is generated by randomly distort the interior vertex of a square mesh. Note that vertex on the boundary $\partial\Omega$ remain untouched.

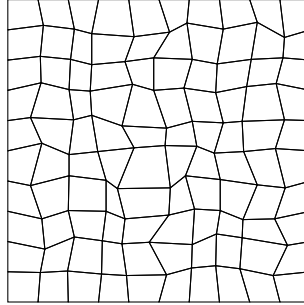


Figure 1.2: An exhibition of a full logically rectangular quasi-uniform quadrilateral mesh $\mathcal{Q}_h$.

1.1.1 Shape Regularity

We give an official definition of uniform shape regularity of a quadrilateral mesh. For any quadrilateral $E \in \mathcal{T}_h$, we can get a set of four triangles T_i , $i \in \{0, 1, 2, 3\}$. We form the triangle T_i by excluding vertex v_i from the quadrilateral E ,

and use the remaining three vertex. We define the following parameters as

$$\begin{aligned} h_E &= \text{diameter of } E, \\ \rho_E &= 2 \min_{1 \leq i \leq 4} \{\text{diameter of largest circle inscribed in } T_i\}. \end{aligned} \quad (1.1)$$

A mesh partition $\mathcal{T}_h$ is considered to be uniformly shape regular if there exists a shape regularity parameter σ_* independent of $\mathcal{T}_h$, such that the ratio

$$\rho_E/h_E \geq \sigma_* > 0, \quad (1.2)$$

for all quadrilaterals $E \in \mathcal{T}_h$.

1.2 Direct Serendipity Space

Serendipity element have the same accuracy of approximation with classical tensor product element while use smaller number of degrees of freedom. However, the classical serendipity space suffer from poor approximation on quadrilaterals. We define local shape functions directly on quadrilaterals according to Arbogast et al. (2022). H^1 and $H(\text{div})$ conforming direct mixed finite element space are constructed with the direct serendipity function space.

To begin with, we introduce some notations and distance auxilliary functions. Let $\mathbb{P}_r(\omega)$ denote the space of polynomials of degree up to r on $\omega \subset \mathbb{R}^d$. The dimension of $\mathbb{P}_r$ can be calculated as

$$\dim \mathbb{P}_r(\mathbb{R}^d) = \binom{r+2}{2} = \frac{(r+2)!}{r!2!}. \quad (1.3)$$

We define the linear polynomial $\lambda_i(\mathbf{x})$ giving the distance of $\mathbf{x} \in \mathbb{R}^2$ to edge e_i as

$$\lambda_i(\mathbf{x}) = -(\mathbf{x} - \mathbf{x}_i^*) \cdot \boldsymbol{\nu}_i, \quad i = 0, 1, 2, 3, \quad (1.4)$$

where $\mathbf{x}_i^* \in e_i$ is any point on the edge e_i . We pick $\mathbf{x}_i^* = \mathbf{x}_{v,i}$ for simplicity. The value of distance function λ_i keeps positive as long as $\mathbf{x}$ is in the interior of the quadrilateral. We define two additional distance functions giving the distance of $\mathbf{x} \in \mathbb{R}^2$ to diagonal

line d_i joining v_i with v_{i+2} , $i \in 0, 1$. Let $\nu_{d,i}$ be the unit normal vector to the diagonal line d_i . The direction of the $\nu_{d,i}$ is defined by counterclockwise rotating of the vector pointing from v_i to v_{i+2} . The resulting diagonal distance function is defined as

$$\lambda_{d,i}(\mathbf{x}) = -(\mathbf{x} - \mathbf{x}_{d,i}^*) \cdot \boldsymbol{\nu}_{d,i}, \quad i = 0, 1, \quad (1.5)$$

where $\mathbf{x}_{d,i}^*$ is any point on the diagonal line d_i . Here, we use the corner point v_i for simplicity, $\mathbf{x}_{d,i}^* = \mathbf{x}_{v,i}$, where $\mathbf{x}_{v,i}$ denotes the coordinates of corner v_i .

Now we present set of nodal basis functions in first and second order direct serendipity space $\mathcal{DS}_1$ and $\mathcal{DS}_2$ defined by Arbogast et al. (2022).

1.2.1 Direct Serendipity Space $\mathcal{DS}_2$

We supplement polynomial space $\mathbb{P}_r(E)$, $r \geq 2$ with two linearly independent functions, $\phi_{s,1}(\mathbf{x})$ and $\phi_{s,2}(\mathbf{x})$, spanning the supplemental function space as $\mathbb{S}_r^{\mathcal{DS}}(E) = \text{span}\{\phi_{s,1}, \phi_{s,2}\}$, $r \geq 2$. We present the second order direct serendipity space as

$$\mathcal{DS}_2(E) = \mathbb{P}_2(E) \oplus \mathbb{S}_2^{\mathcal{DS}}(E). \quad (1.6)$$

The supplemental functions are defined as

$$\phi_{s,i}(\mathbf{x}) = \lambda_{i+1}(\mathbf{x})\lambda_{i+2}(\mathbf{x})R_{i,i+2}(\mathbf{x}), \quad i \in \{0, 1\}, \quad (1.7)$$

where $i \equiv i \pmod{4}$. We use a simple definition for rational function $R_{i,i+2}$ as

$$R_{i,i+2}(\mathbf{x}) = \frac{\lambda_i(\mathbf{x}) - \lambda_{i+2}(\mathbf{x})}{\lambda_i(\mathbf{x}) + \lambda_{i+2}(\mathbf{x})}, \quad (1.8)$$

which satisfies the conditions as

$$R_{i,i+2}|_{e_i} = -1, \quad R_{i,i+2}|_{e_{i+2}} = 1. \quad (1.9)$$

We continue to define

$$R_i(\mathbf{x}) = \frac{1}{2} \left(1 - R_{i,i+2}(\mathbf{x}) \right) = \frac{\lambda_{i+2}(\mathbf{x})}{\lambda_i(\mathbf{x}) + \lambda_{i+2}(\mathbf{x})}, \quad i \in \{0, 1, 2, 3\} \quad (1.10)$$

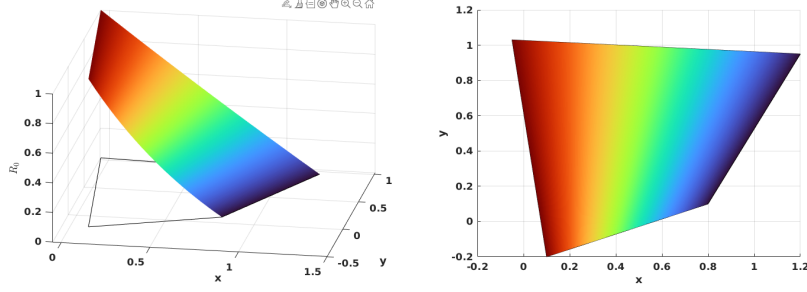


Figure 1.3: Rational function R_0 .

where $i \equiv i \pmod{4}$ as usual, and $R_{i,i+2} = -R_{i+2,i}$ according to the equation (1.8). Here, $R_i|_{e_i} = 1$, and $R_i|_{e_{i+2}} = 0$ by definition. In Figure 1.3, we draw R_0 on a general quadrilateral element, which evaluates 1 on the edge e_0 , and evaluates 0 on the opposite edge e_2 .

The edge nodal basis functions for $\mathcal{DS}_2$ are constructed as

$$\varphi_{e,i}^{(2)}(\mathbf{x}) = \frac{\lambda_{i+1}(\mathbf{x})\lambda_{i+3}(\mathbf{x})R_i(\mathbf{x})}{\lambda_{i+1}(\mathbf{x}_{e,i})\lambda_{i+3}(\mathbf{x}_{e,i})R_i(\mathbf{x}_{e,i})}, \quad (1.11)$$

where $\mathbf{x}_{e,i}$ is the coordinates of the middle point of the edge e_i . Here, $i \equiv i \pmod{4}$. We can restrict the edge basis function to edges as

$$\varphi_{e,i}^{(2)}|_{e,j} = \begin{cases} \frac{((\mathbf{x} - \mathbf{x}_{v_{i+1}}) \cdot \boldsymbol{\nu}_i)((\mathbf{x} - \mathbf{x}_{v_{i+3}}) \cdot \boldsymbol{\nu}_i)}{|e_i|^2/4}, & i = j \\ 0, & i \neq j \end{cases} \quad (1.12)$$

In Figure 1.4, we present $\varphi_{e,0}^{(2)}$ on a general quadrilateral element, where $\varphi_{e,0}^{(2)}$ evaluates 0 on the edges $\{e_i, \quad i \neq 0\}$, and evaluates a quadratic bubble function on the edge e_0 .

The vertex nodal basis functions for $\mathcal{DS}_2$ are constructed as

$$\varphi_{v,i}^2(\mathbf{x}) = \frac{\lambda_{i+2}(\mathbf{x})\lambda_{i+3}(\mathbf{x}) - \sum_{k=i}^{i+1} \lambda_{i+2}(\mathbf{x}_{e,i})\lambda_{i+3}(\mathbf{x}_{e,i})\varphi_{e,i}^2(\mathbf{x})}{\lambda_{i+2}(\mathbf{x}_{v,i})\lambda_{i+3}(\mathbf{x}_{v,i}) - \sum_{k=i}^{i+1} \lambda_{i+2}(\mathbf{x}_{e,i})\lambda_{i+3}(\mathbf{x}_{e,i})\varphi_{e,i}^2(\mathbf{x}_{v,i})}, \quad (1.13)$$

where $\mathbf{x}_{v,i}$ is the coordinates of the vertex node v_i . The i index is understood in the sense of module of 4 as well. In Figure 1.5, we show $\varphi_{v,0}^2$ on a general quadrilateral.

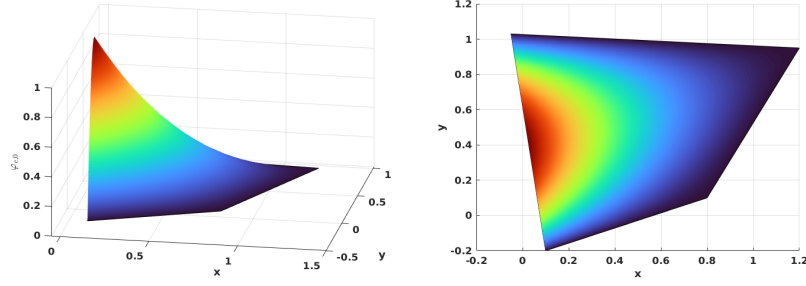


Figure 1.4: Edge nodal basis function $\varphi_{e,0}^2$ in $\mathcal{DS}_2$.

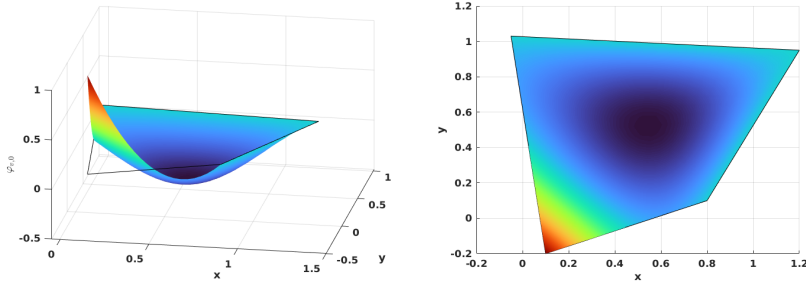


Figure 1.5: Vertex nodal basis function $\varphi_{v,0}^2$ in $\mathcal{DS}_2$.

We conclude that unisolvence follows directly from the constructed nodal basis functions.

1.2.2 Direct Serendipity Space $\mathcal{DS}_1$

The supplemental space for $\mathcal{DS}_1$ differs from that for $\mathcal{DS}_2$. We present the first order serendipity as

$$\mathcal{DS}_1(E) = \mathbb{P}_1(E) \oplus \text{span}\{R\}, \quad (1.14)$$

where $R(\mathbf{x}_{v,0}) = R(\mathbf{x}_{v,2}) = -R(\mathbf{x}_{v,1}) = -R(\mathbf{x}_{v,3}) = 1$. The construction of rational function R is not unique. We present one way to construct $R(\mathbf{x})$ with edge nodal basis function (1.11) in $\mathcal{DS}_2$ as

$$R(\mathbf{x}) = r(\mathbf{x}) - \sum_{i=0}^3 r(\mathbf{x}_{e,i}) \varphi_{e,i}^2(\mathbf{x}), \quad (1.15)$$

where auxiliary function $r(\mathbf{x})$ is defined as

$$r(\mathbf{x}) = \frac{\lambda_2(\mathbf{x})\lambda_3(\mathbf{x})}{\lambda_2(\mathbf{x}_{v,0})\lambda_3(\mathbf{x}_{v,0})} - \frac{\lambda_0(\mathbf{x})\lambda_3(\mathbf{x})}{\lambda_0(\mathbf{x}_{v,1})\lambda_3(\mathbf{x}_{v,1})} + \frac{\lambda_0(\mathbf{x})\lambda_1(\mathbf{x})}{\lambda_0(\mathbf{x}_{v,2})\lambda_1(\mathbf{x}_{v,2})} - \frac{\lambda_1(\mathbf{x})\lambda_2(\mathbf{x})}{\lambda_1(\mathbf{x}_{v,3})\lambda_2(\mathbf{x}_{v,3})}. \quad (1.16)$$

In Figure 1.6, we plot the rational function on a general quadrilateral.

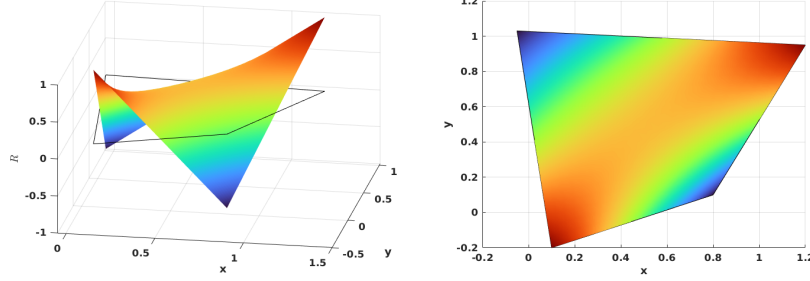


Figure 1.6: Rational function in $\mathcal{DS}_1$.

We define vertex nodal basis function accordingly as

$$\varphi_{v,i}^1(\mathbf{x}) = \frac{\lambda_{d,m}(\mathbf{x}) - \frac{1}{2}\lambda_{d,m}(\mathbf{x}_{v,i+2})(1 + (-1)^i R(\mathbf{x}))}{\lambda_{d,m}(\mathbf{x}_{v,i}) - \lambda_{d,m}(\mathbf{x}_{v,i+2})}, \quad (1.17)$$

where $m \equiv i + 1 \pmod{2}$, and index i is understood in the sense of modulo of 4. In Figure 1.7, we show the vertex nodal basis function $\varphi_{v,0}^1$ in $\mathcal{DS}_1$ on a general quadrilateral, which evaluates 1 at vertex v_0 and 0 at other vertex.

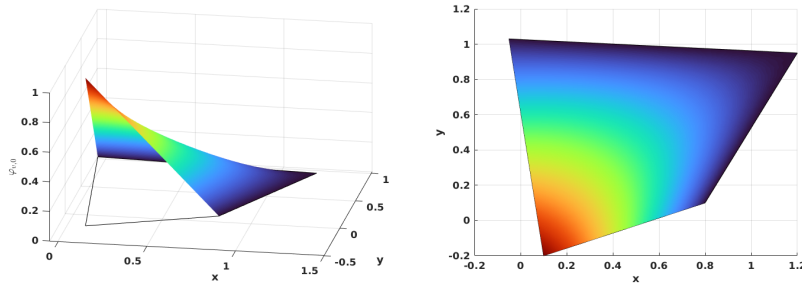


Figure 1.7: Vertex nodal basis function in $\mathcal{DS}_1$.

We first define an interpolation operator of function $v \in L^2(\Omega)$ onto $\mathcal{DS}_r(K_i)$, $r = 1, 2$. Here, K_i has different meanings with different types of basis functions.

For each basis functions associated with edge node, we set $K_i = e_i$. For each basis functions associated with vertex node, we set K_i being a set of edges sharing the vertex v_i . Following the work by Arbogast et al. (2022), we define a interpolation operator $\mathcal{J}_h^r : W_p^l(E) \rightarrow \mathcal{DS}_r$ as

$$\mathcal{J}_h^r v(\mathbf{x}) = \sum_{i=1}^{N_r} \varphi_i(\mathbf{x}) \int_{K_i} \psi_i(\mathbf{y}) v(\mathbf{y}) d\mathbf{y} \in \mathcal{DS}_r, \quad (1.18)$$

where $1 \leq p \leq \infty$ and $l > 1/p$ (but $l \geq 1$ if $p = 1$), and $\psi(\mathbf{x})$ is the dual nodal basis such that

$$\int_{K_i} \psi_i(\mathbf{x}) \varphi_j(\mathbf{x}) d\mathbf{x} = \delta_{i,j}, \quad i, j \in \{0, 1, 2, \dots, N_r\}, \quad (1.19)$$

where N_r is the number of nodal basis defined on K_i . We present the following two lemmas regarding the constructed interpolation operator for future reference.

Lemma 1.1. (*Boundedness of the interpolation operator.*)

Let $\mathcal{T}_h$ be uniformly shape regular (1.2) with shape regularity parameter σ_* . For each element $E \in \mathcal{T}_h$, There exists a bounded interpolation operator (1.18). For every $E \in \mathcal{T}_h$, suppose that the basis functions of $\mathcal{DS}_r(E)$ are constructed using λ_{24} and λ_{13} such that the zero set $\mathcal{Z}_{i,i+2}$ intersects e_{i+3} and e_{i+1} . Moreover, assume that $R_{i,i+2}$ is uniformly differentiable functions of the vertices of E up to order m . Then for $r \geq 1$, $E \in \mathcal{T}_h$, $1 \leq q \leq \infty$, and any nonnegative integer m ,

$$\|\mathcal{J}_h^r v\|_{W_q^m} \leq C(\sigma_*, m, q) \sum_{k=0}^l h_E^{k-m+\frac{2}{q}-\frac{2}{p}} |v|_{W_p^k(E^*)}, \quad (1.20)$$

where $E^* = \cup_{F \in \mathcal{T}_h, F \cap E \neq \emptyset}$ and $|\cdot|_{W_p^k}$ is the seminorm of k -th order derivatives.

Proof. See Arbogast et al. (2022). □

Lemma 1.2. (*Approximability of the interpolation operator.*)

Under the assumptions of 1.1, there exists a constant $C = C(r, \sigma_*) > 0$ such that for all functions $v \in W_{l,p}(E^*)$, with $1 \leq p \leq \infty$ and $l > 1/p$ or ($l \geq 1$ if $p = 1$),

$$\|v - \mathcal{J}_h^r v\|_{W_p^m} \leq C h_E^{l-m} |v|_{W_p^l(E^*)}, \quad 0 \leq m \leq \min(l, r+1). \quad (1.21)$$

Moreover, there exists a constant $C = C(r, \sigma_*) > 0$, independent of $h = \max_{E \in \mathcal{T}}$, such that for all functions $v \in W_p^l(\Omega)$,

$$\left(\sum_{E \in \mathcal{T}_h} \|v - \mathcal{I}_h^r v\|_{W_p^m}^p \right)^{1/p} \leq Ch^{l-m} |v|_{W_p^l(\Omega)}, \quad 0 \leq m \leq \min(l, r+1). \quad (1.22)$$

Proof. See Arbogast et al. (2022). □

1.3 Mixed Finite Element

We introduce a H^1 and a $H(\text{div})$ conforming mixed element space constructed with the pre-defined direct serendipity space.

Recall Ciarlet's definition Ciarlet (2002) of a finite element.

Definition 1.1. A triple $(E, X(E), \varphi_j)$ defines a finite element.

- Element $E \in \mathcal{T}_h$;
- Space of shape function $X(E)$, and a set of shape function $\varphi_i \in X(E)$. The dimension of the shape function space $\dim(X(E)) = n$;
- A set of continuous and linear functionals φ_j

$$\phi_j : X(E) \supset \mathcal{X}(E) \rightarrow \mathbb{R}, \quad (1.23)$$

such that unisolvence condition is achieved as

$$\langle \phi_j, \varphi_i \rangle = \delta_{ij}, \quad i, j \in \{1, 2, 3, \dots, n\} \quad (1.24)$$

In the following sections, we state our shape function space constructed using first and second order direct serendipity space and corresponding degrees of freedom. We also provide explicit construction of shape functions.

1.3.1 $H(\text{div})$ Conforming Element

We have some choice for $H(\text{div})$ confirming space. For example, we can choose the famous Ravier-Thomas element by Raviart and Thomas (1977). The serendipity space is the precursor of the Brezzi-Douglas-Marini element by Brezzi et al. (1985). We recover Arbogast-Correa element by Arbogast and Correa (2016) if we use mapped direct serendipity space. We generated a reduced $H(\text{div})$ approximation, that $\nabla \cdot \mathbf{u}$ is approximated one less order than approximation of $\mathbf{u}$.

We write the rotated 2D exact sequence for energy space as

$$H^1 \xrightarrow{\text{curl}} H(\text{div}) \xrightarrow{\text{div}} L^2 \quad (1.25)$$

$$\cup \quad \quad \cup \quad \quad \cup \quad (1.26)$$

$$\mathcal{DS}_2 \xrightarrow{\text{curl}} \mathbf{V}_r^{r-1} \xrightarrow{\text{div}} \mathbb{P}_{r-1}. \quad (1.27)$$

We define the first order reduced $H(\text{div})$ conforming space with $\mathcal{DS}_2$ as

$$\mathbb{V}_1^0 = \mathbb{P}_1^2 \oplus \text{curl}\mathcal{S}_2^{\mathcal{DS}}. \quad (1.28)$$

The reduced space $\mathbb{V}_1^0$ can be given a global basis with the normal components of the shape functions merged continuously on the shared edge of two elements. The dimension of the derived space is

$$\dim(\mathbb{V}_1^0) = 2 \times 3 + 2 = 8. \quad (1.29)$$

We need two independent basis functions on each edge of a quadrilateral E , one of which has a vanishing average flux on the edge and the other one has a non-vanishing average flux on the edge.

We state the degree of freedom of element as normal flux on each edge as

$$\int_{e_i} \mathbf{u} \cdot \boldsymbol{\nu} \varphi_i d\mathbf{x}, \quad \varphi_i \in \mathbb{V}_1^0. \quad (1.30)$$

We need two independent shape functions on each edge. We present a construction of two independent basis functions on each edge: $\varphi_{l,i}$ and $\varphi_{c,i}$, such that

$$\varphi_{c,i} \cdot \boldsymbol{\nu}_j|_{e_j} = \begin{cases} 0, & j \neq i, \\ 1, & j = i. \end{cases} \quad (1.31)$$

$$\boldsymbol{\varphi}_{l,i} \cdot \boldsymbol{\nu}_j|_{e_j} = \begin{cases} 0, & j \neq i, \\ l_i(\mathbf{x}), & j = i, \end{cases} \quad (1.32)$$

where l_i is a non vanishing linear polynomial defined on edge e_i orthogonal to one, i.e., $\int_{e_i} l_i ds = 0$.

To begin with, we write curl of the auxiliary rational function R_i (1.8) as

$$\text{curl} R_i(\mathbf{x}) = \frac{\boldsymbol{\tau}_{i+2}(\mathbf{x})\lambda_i(\mathbf{x}) - \boldsymbol{\tau}_i(\mathbf{x})\lambda_{i+2}(\mathbf{x})}{(\lambda_i(\mathbf{x}) + \lambda_{i+2}(\mathbf{x}))^2}, \quad (1.33)$$

where the index i is understood with modulo of 4. We first present a construction of basis function $\boldsymbol{\varphi}_{l,i}$ as

$$\tilde{\boldsymbol{\varphi}}_{l,i} = \text{curl}(\lambda_{i+1}\lambda_{i+3}R_i) = (\boldsymbol{\tau}_{i+1}\lambda_{i+3} + \lambda_{i+1}\boldsymbol{\tau}_{i+3})R_i + \lambda_{i+1}\lambda_{i+3}\text{curl}R_i. \quad (1.34)$$

We then normalize it such that l_i varying from -1 to 1 as

$$\boldsymbol{\varphi}_{l,i} = \frac{\tilde{\boldsymbol{\varphi}}_{l,i}(\mathbf{x})|e_i|}{4\lambda_{i+1}(\mathbf{x}_{e,i})\lambda_{i+3}(\mathbf{x}_{e,i})R_i(\mathbf{x}_{e,i})}, \quad (1.35)$$

where the index i is understood in the sense of modulo 4, and edge length $|e_i| = |\mathbf{x}_{v,i} - \mathbf{x}_{v,i+3}|$. In Figure 1.8, we draw $\boldsymbol{\varphi}_{l,2}$ on the left element and $\boldsymbol{\varphi}_{l,0}$ on the right element sharing a edge, and show normal flux joining continuously as a linear function on the shared edge.

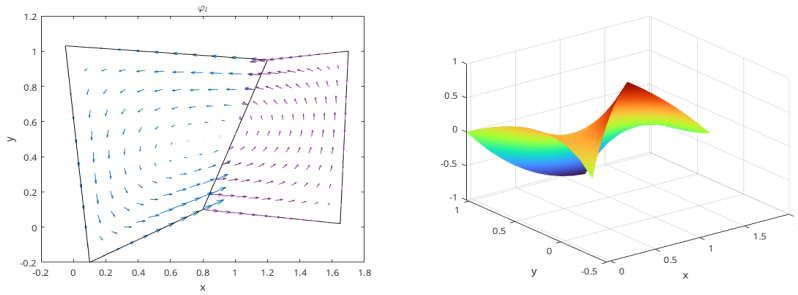


Figure 1.8: Linear basis function $\boldsymbol{\varphi}_l$.

The construction of a the constant basis function is a bit more complicated. We follow the process by Arbogast et al. (2022). We start by defining a linear nodal

function $\phi_{v,i}^* \in \mathcal{DS}_2(E)$ such that $\phi_{v,i}^*(\mathbf{x}_{v,k}) = \delta_{ik}$. Its restriction to each edge of E is a linear function as

$$\phi_{v,i}^*(\mathbf{x}) = \varphi_{v,i}^2(\mathbf{x}) + \left[\frac{1}{2}\varphi_{e,i}^2 + \frac{1}{2}\varphi_{e,i+1}^2 \right], \quad (1.36)$$

where the index i is understood in modulo of 4. Then define $\boldsymbol{\varphi}_{v,i}^*(\mathbf{x}) = \text{curl}\phi_{v,i}^*$ so that

$$\boldsymbol{\varphi}_{v,i}^* \cdot \boldsymbol{\nu}_j|_{e,j} = \begin{cases} 1/|e_i|, & j = i, \\ -1/|e_i|, & j = i + 1, \\ 0, & j = i + 2, i + 3. \end{cases} \quad (1.37)$$

We then define the "constant" basis function defined on the edge as

$$\boldsymbol{\varphi}_{e,i} = \frac{(\mathbf{x}_{v,i} - \mathbf{x}_{v,i+2}) \cdot \boldsymbol{\nu}_{i+1}|e_{i+1}|\boldsymbol{\varphi}_{v,i}^* + \mathbf{x} - \mathbf{x}_{v,i+2}}{(\mathbf{x}_{v,i} - \mathbf{x}_{v,i+2}) \cdot \boldsymbol{\nu}_{i+1}|e_{i+1}|/|e_i| + (\mathbf{x}_{v,i} - \mathbf{x}_{v,i+2}) \cdot \boldsymbol{\nu}_i}, \quad (1.38)$$

where the index i is again understood in the sense of modulo of 4. The defined basis function $\boldsymbol{\varphi}_{e,i}$ has flux of constant value 1 on e_i and 0 on all the other edges. In Figure 1.9, we draw $\boldsymbol{\varphi}_{c,2}$ on the left element and $\boldsymbol{\varphi}_{c,0}$ on the right element sharing a edge, and show normal flux joining continuously as a constant 1 on the shared edge.

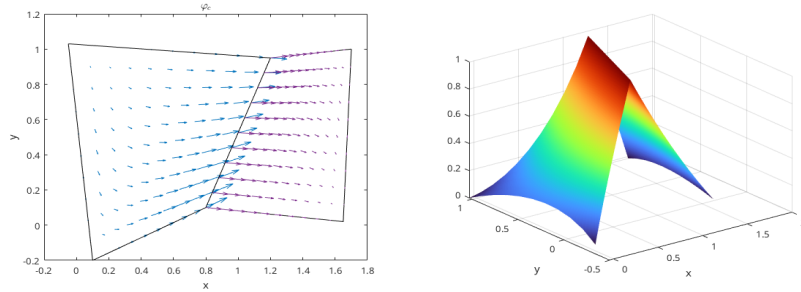


Figure 1.9: Constant basis function $\boldsymbol{\varphi}_c$.

The accuracy and stability of such element has been well studied and tested in the work by Arbogast et al. (2022). We state the result directly for completeness. Consider a projection operator π_D preserves element average divergence and average edge normal fluxes.

$$\pi_D : H(\text{div}; \Omega) \cap (L^{2+\epsilon}(\Omega))^2 \rightarrow \mathbb{V}_1^0, \quad (1.39)$$

where $\epsilon > 0$. The global projection operator π is understood as combination of local operator π_E according to degree of freedom. The operator π also satisfies the commuting diagram property as

$$\mathcal{P}_{W_s} \nabla \cdot \mathbf{v} = \nabla \cdot \pi \mathbf{v}, \quad (1.40)$$

where $\mathcal{P}_{W_s}$ is the L^2 -orthogonal projection operator onto $W_s = \nabla \cdot \mathbb{V}_1^0$, which is a constant value in our case.

Lemma 1.3. *Stability and approximation properties.*

Under the assumption of 1.1, there exists a constant $C > 0$, independent of $\mathcal{T}_h$ and $h > 0$, such that

$$\|p - p_h\|_{m,\Omega} \leq Ch^{l+1-m} |p|_{l+1,\Omega}, \quad l = 0, 1, 2, \dots, r, \quad (1.41)$$

where $p_h \in \mathcal{DS}_2$. We have

$$\begin{aligned} \|\mathbf{u} - \pi_D \mathbf{u}\|_{0,\Omega} &\leq C \|\mathbf{u}\|_{k,\Omega} h^k, & k = 1, \dots, 2 \\ \|\nabla \cdot (\mathbf{u} - \pi_D \mathbf{u})\|_{0,\Omega} &\leq C \|\nabla \mathbf{u}\|_{k,\Omega} h^k, & k = 0, \dots, 1 \\ \|p - \mathcal{P}_{W_s} p\|_{0,\Omega} &\leq C \|p\|_{k,\Omega} h^k, & k = 0, \dots, 1 \end{aligned} \quad (1.42)$$

Moreover, the discrete inf-sup condition holds

$$\sup_{\mathbf{v}_h \in \mathbb{V}_1^0} \frac{(\omega_h, \nabla \cdot \mathbf{v}_h)}{\|\mathbf{v}_h\|_{H(\text{div})}} \geq \gamma \|\omega_h\|_{0,\Omega}, \quad \forall \omega_h \in W_s, \quad (1.43)$$

holds for some $\gamma > 0$ independent of $\mathcal{T}_h$ and $h > 0$.

Proof. See Arbogast et al. (2022). □

1.3.2 H^1 Conforming Element

Stable mixed finite element pair for Stokes problem has been widely researched. We are referring to the element discussed by Christine Bernardi (1985) and by Fortin (1981). A more general type of element was discussed by Arbogast and Wheeler (2005).

Vertex basis mapping to sobolev space difficulty. Need Clément interpolation. Define Π operator two parts.

We consider the following space for each element $E \in \mathcal{T}_h$ as

$$V_1^{\mathcal{BR}}(E) = (\mathcal{DS}_1(E))^2 \oplus \text{span}\{\mathbf{b}_0, \mathbf{b}_1, \mathbf{b}_2, \mathbf{b}_3\}, \quad (1.44)$$

where $\mathbf{b}_i$ is a quadratic bubble function defined on each edge e_i . We pick previously defined function $\varphi_{e,i} \in \mathcal{DS}_2$ as

$$\mathbf{b}_i = \varphi_{e,i}(\mathbf{x})\mathbf{n}_i \quad (1.45)$$

which is quadratic when restricted to edge e_i .

Lemma 1.4. *For an element $E \in \mathcal{T}_h$, $\mathbf{u}_h$ is uniquely defined by the following degrees of freedom:*

(i) *for corner point v_i and Cartesian direction unitnormal vectors $\{\mathbf{i}, \mathbf{j}\}$,*

$$DOF_{(v_i, \mathbf{i})}^{(i)}(\mathbf{u}_h) = \mathbf{u}_h(v_i) \cdot \mathbf{i}, \quad DOF_{(v_i, \mathbf{j})}^{(i)}(\mathbf{u}_h) = \mathbf{u}_h(v_i) \cdot \mathbf{j};$$

(ii) *for edge e_i ,*

$$DOF_{e_i}^{(ii)} = \langle \mathbf{u}_h \cdot \boldsymbol{\nu}_i, 1 \rangle,$$

Proof. We prove the lemma by construction. We pick nodal basis on vertex v_i in two Cartesian direction as

$$N_{v_i, x} = \begin{bmatrix} \psi_{v,i}(\mathbf{x}) \\ 0 \end{bmatrix}, \quad N_{v_i, y} = \begin{bmatrix} 0 \\ \psi_{v,i}(\mathbf{x}) \end{bmatrix} \quad (1.46)$$

We pick edge basis on edge e_i as

$$N_e = \varphi_{e,i}\boldsymbol{\nu}_i \quad (1.47)$$

Note that bubble functions are quadratic when restricted to edge e_i , while nodal basis functions are linear when restricted to edge e_i , therefore bubble functions cannot be in the same space as nodal basis function. We conclude that this element is well-defined, i.e., the degrees of freedom uniquely determine the function. $\square$

We need a π operator, which mappes $(H^1(E))^2$ onto $V_1^{\mathcal{BR}}$ for our spaces. For $\mathbf{v} \in (H^1(\Omega))^2$, let $\pi\mathbf{v} \in \mathbb{V}_1^{\mathcal{BR}}$ be defined as the interpolant of the degrees of freedom from 1.4. For each F , we require the following as

- (i) For each corner point $v \in \partial F$

$$\pi\mathbf{v}(\mathbf{x}_v) \cdot \mathbf{i} = \mathcal{J}_{\mathcal{DS}_1} \mathbf{v}(\mathbf{x}_v) \cdot \mathbf{j}. \quad (1.48)$$

- (ii) On each edge $e \subset \partial E$,

$$\langle \pi\mathbf{u} \cdot \boldsymbol{\nu}, \lambda_0 \rangle = \langle \mathbf{u} \cdot \boldsymbol{\nu}, \lambda_0 \rangle_e \quad \forall \lambda_0 \in P_0(e). \quad (1.49)$$

The interpolation operator $\mathcal{I}_{\mathcal{DS}_1}$ has been defined in 1.18. We claim that we can represent for $\mathbf{u} \in (H^1(\Omega))^2$,

$$\pi\mathbf{u} = \mathcal{I}_{\mathcal{DS}_1} \mathbf{u} + \sum_{i=0}^3 \alpha_i \mathbf{b}_i, \quad (1.50)$$

where the coefficient $\alpha_i = \left(\int_{e_i} (\mathbf{u} - \mathcal{J}_{\mathcal{DS}_1} \mathbf{u}) \cdot \boldsymbol{\nu}_i d\sigma \right) / \int_{e_i} \mathbf{b}_i \cdot \boldsymbol{\nu}_i d\sigma$. It is obvious that $\pi\mathbf{u} \in \mathbb{V}_1^{\mathcal{BR}}$. We now prove the approximability and stability of our projection operator and eventually derive the inf – sup condition.

Lemma 1.5. *Assuming that $u \in (H^1(\Omega))^2$.*

- (i) *there exists a bounded linear operator π*

- (ii) *$\forall \mathbf{v} \in (H^1(\Omega))^2$*

1.3.3 Test Stokes Problem

We test our modified Bernardi-Raugel elements on the incompressible isotropic elasticity problem with a simple homogenous Dirichlet boundary condition:

Find the displacement $\mathbf{u}$ and pressure p such that

$$\begin{aligned} -\Delta \mathbf{u} + \nabla p &= \mathbf{f} \quad \text{in } \Omega, \\ \nabla \cdot \mathbf{u} &= 0 \quad \text{in } \Omega, \\ \mathbf{u} &= \mathbf{g} \quad \text{on } \partial\Omega. \end{aligned} \quad (1.51)$$

We define a simple choice of energy space

$$\begin{aligned}\mathbb{V} &= \{\mathbf{v} \in (H^1\Omega)^2 : \mathbf{v} = \mathbf{0} \text{ on } \partial\Omega\}, \\ \mathbb{U} &= \{\mathbf{u} \in (H^1\Omega)^2 : \mathbf{u} = \mathbf{g} \text{ on } \partial\Omega\} = \tilde{\mathbf{g}} + \mathbb{V}, \\ \mathbb{W} &= \{p \in L^2(\Omega)/\mathbb{R}\},\end{aligned}\tag{1.52}$$

where $\tilde{\mathbf{g}}$ is a finite energy lift. We assume that $\mathbf{g} \in (H^{1/2}(\Omega))^d$ and understand as a continuous extension $\tilde{\mathbf{g}} \in (H^1(\Omega))^d$ from the boundary into the domain. We write the problem in weak form as:

Find $\mathbf{u} \in \mathbb{U}$ and $p \in L^2(\Omega)$ such that

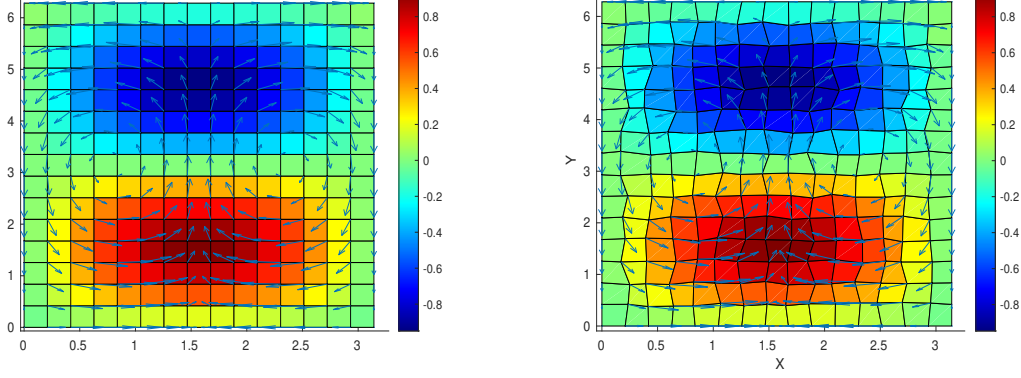
$$\begin{aligned}(\nabla \mathbf{u}, \nabla \mathbf{v}) - (p, \nabla \cdot \mathbf{v}) &= (\mathbf{f}, \mathbf{v}), \quad \forall \mathbf{v} \in \mathbb{V}, \\ (\nabla \cdot \mathbf{u}, \omega) &= 0, \quad \forall \omega \in L^2(\Omega).\end{aligned}\tag{1.53}$$

We consider the true solution of test problem (1.53) defined on a square region $\Omega = [0, \pi] \times [0, 2\pi]$. We manufacture a displacement and pressure as

$$\mathbf{u} = \begin{bmatrix} \cos(x) \sin(y) \\ -\sin(x) \cos(y) \end{bmatrix}, \quad \nabla p = \begin{bmatrix} \cos(x) \sin(y) \\ \sin(x) \cos(y) \end{bmatrix},\tag{1.54}$$

which has a 0 normal flux on the boundary. We deal with the constant kernel by enforcing a condition that $\int_{\Omega} p d\mathbf{x} = 0$ and solve the linear system directly. The numerical results are computed on rectangular 1.10a and logically rectangular quadrilateral 1.10b meshes. We calculate the L^2 norm of the residual (i.e., the difference) of the pressure, velocity, and derivative of velocity.

In Table 1.1 and Table 1.2 we display the result of a calculation of convergence on $n \times n$ rectangular and distorted grids, respectively, for $n = 8, 16, 32, 64$. Second order convergence rate is achieved by velocity and its corresponding derivatives. A first order convergence rate is achieved by pressure. Both meet our expectations for optimal convergence order.



(a) Rectangular Grids. Shown are pressure (color) and velocity (arrows). (b) Distorted Grids. Shown are pressure (color) and velocity (arrows).

Grid size	8×8	16×16	32×32	64×64
$L^2(\text{res}_u)$	4.502E-1	1.052E-1	2.513E-2	6.130E-3
$L^2(\text{res}_p)$	1.039E+0	4.463E-1	2.093E-1	1.019E-1
$L^2(\text{res}_{du})$	2.632E+0	5.845E-1	1.374E-1	3.330E-2

Table 1.1: Rectangular Grids. L^2 -error in the residual of the solution u (res_u), the pressure p (res_p), and the of derivatives of the solution Du (res_{du}).

Grid size	8×8	16×16	32×32	64×64
$L^2(\text{res}_u)$	4.603E-1	1.074E-2	2.567E-2	6.266E-3
$L^2(\text{res}_p)$	1.070E+0	4.670E-1	2.207E-1	1.078E-1
$L^2(\text{res}_{du})$	2.685E+0	5.981E-1	1.525E-1	4.580E-2

Table 1.2: Distorted Grids. L^2 -error in the residual of the solution u (res_u), the pressure p (res_p), and the of derivatives of the solution Du (res_{du}).

Works Cited

Todd Arbogast and Maicon R. Correa. Two families of $\mathcal{H}(\text{div})$ mixed finite elements on quadrilaterals of minimal dimension. *SIAM Journal on Numerical Analysis*, 54(6):3332–3356, 2016. doi: 10.1137/15M1013705. URL <https://doi.org/10.1137/15M1013705>.

Todd Arbogast and Mary F. Wheeler. A family of rectangular mixed elements with a continuous flux for second order elliptic problems. *SIAM Journal on Numerical Analysis*, 42(5):1914–1931, 2005. doi: 10.1137/S0036142903435247. URL <https://doi.org/10.1137/S0036142903435247>.

Todd Arbogast, Zhen Tao, and Chuning Wang. Direct serendipity and mixed finite elements on convex quadrilaterals. *Numerische Mathematik*, 150:929–974, 2022.

Franco Brezzi, Jim Douglas Jr., and L. D. Marini. Two families of mixed finite element for second order elliptic problems. *Numerische Mathematik*, 47:217–235, 1985.

Genevieve Raugel Christine Bernardi. Analysis of some finite elements for the stokes problem. *Math. of Comput.*, 44:71–79, 1985.

Philippe G. Ciarlet. *The Finite Element Method for Elliptic Problems*. Society for Industrial and Applied Mathematics, 2002. doi: 10.1137/1.9780898719208. URL <https://epubs.siam.org/doi/abs/10.1137/1.9780898719208>.

Michel Fortin. Old and new finite elements for incompressible flows. *International Journal for Numerical Methods in Fluids*, 1:347–364, 1981. URL <https://api.semanticscholar.org/CorpusID:122770101>.

P. A. Raviart and J. M. Thomas. A mixed finite element method for 2-nd order elliptic problems. In Ilio Galligani and Enrico Magenes, editors, *Mathematical Aspects of Finite Element Methods*, pages 292–315, Berlin, Heidelberg, 1977. Springer Berlin Heidelberg. ISBN 978-3-540-37158-8.